

UNIVERSITI KEBANGSAAN MALAYSIA
The National University of Malaysia

Compound Semiconductors

KKKC6353 Nanoelectronic Devices — Lecture 4, Document 4 of 4

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October 2026

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The argument of this document

> **Recap from earlier lectures**
>
> Lecture 1: at high fields electrons in silicon saturate at about 10^7 cm/s. Lecture 2: quantum wells make lasers and LEDs, and the HEMT puts electrons in an "express lane" away from their donors. Lecture 3: tunnel junctions join the cells of multi-junction solar cells.
>
> This document gives the materials physics behind those devices, and adds HBTs and power devices.

Silicon is not the best semiconductor at anything except being silicon. It has a native oxide of superb quality, it can be grown as near-perfect crystals 300 mm across, and sixty years of investment have made it the cheapest way to build billions of transistors. On almost every other measure, some **compound semiconductor** does better: faster electrons (GaAs, InGaAs, InP), light emission (every LED and laser), higher breakdown fields (SiC, GaN), or higher operating temperature.

This document explains where those advantages come from and how devices exploit them. It makes five points:

1. Compounds of group III and group V elements (and others) can be **mixed as alloys**, so the band gap can be chosen continuously, and different materials can be grown on each other as **heterostructures**. This freedom is the single most important difference from silicon.
2. Many of these materials have a **direct gap** and a **light electron**. The first makes them emit light; the second gives high mobility and, through the transferred-electron effect, a velocity peak above silicon's.
3. Heterojunctions make the two compound-semiconductor transistors that matter: the **high-electron-mobility transistor (HEMT)** and the **heterojunction bipolar transistor (HBT)**. In gallium nitride the HEMT needs no doping at all: polarisation supplies the electrons.
4. **Wide-gap** materials (SiC, GaN) have critical fields ten times silicon's. The ideal resistance of a power device falls as the *cube* of the critical field, which is why electric cars and phone chargers now use them.
5. Compound semiconductors have **not** replaced silicon in logic, and the reasons, including the density-of-states bottleneck, are as instructive as their successes.

1 The compound semiconductor family

> **Key ideas**
>
> III–V semiconductors (GaAs, InP, GaN, …) pair a group III and a group V atom; II–VI (ZnSe, CdTe) and IV–IV (SiC) compounds are also semiconductors.
>
> Alloys such as $Al_xGa_{1-x}As$ and $In_xGa_{1-x}As$ let the band gap and lattice constant be chosen continuously.
>
> Layers with matched (or nearly matched) lattice constants can be grown on each other without defects: heterostructures.

1.1 Structure and bonding

Most III–V compounds crystallise in the **zinc-blende** structure: the diamond lattice of silicon with alternate atoms replaced by the group III and group V element. Each atom still has four tetrahedral bonds, but the bonds are partly ionic, because the group V atom attracts the shared electrons more strongly. The nitrides (GaN, AlN, InN), SiC polytypes such as 4H-SiC, and ZnO crystallise in the hexagonal **wurtzite** structure. This matters for GaN, because a wurtzite crystal has no centre of symmetry and can be electrically **polarised** (Section 3.4).

1.2 The properties that matter

Property (300 K)	Si	GaAs	InP	$In_{0.53}Ga_{0.47}As$	4H-SiC	GaN
Band gap (eV)	1.12 (ind.)	1.42 (dir.)	1.34 (dir.)	0.74 (dir.)	3.26 (ind.)	3.4 (dir.)
Electron effective mass (m_0)	0.19 / 0.98	0.067	0.08	0.041	≈ 0.3–0.4	0.2
Electron mobility ($cm^2/V\,s$)	1,400	8,500	5,400	≈ 12,000	≈ 900	≈ 1,000 (2DEG ≈ 2,000)
Peak (saturated) velocity (10^7 cm/s)	1.0 (1.0)	2 (1.0)	2.5 (1.0)	≈ 2.5 (0.8)	2.0 (2.0)	≈ 2.5–3 (1.5–2)
Critical field (MV/cm)	0.3	0.4	0.5	≈ 0.2	2.5–3	3.3
Relative permittivity	11.7	12.9	12.5	13.9	9.7	9.5
Thermal conductivity (W/cm K)	1.5	0.5	0.7	0.05	3.7–4.9	1.3–2.3
Lattice constant a (Å)	5.431	5.653	5.869	5.869	3.073	3.189

Typical values from the Ioffe Institute database and the literature; mobilities are for undoped material. The CSUN deck in the old notes has a similar table.

Each column tells a story. GaAs and InGaAs have light electrons and high mobility: fast transistors at low fields. InP is the substrate on which $In_{0.53}Ga_{0.47}As$ is lattice-matched, and the material of optical-fibre lasers. SiC and GaN have wide gaps and critical fields ten times silicon's: power devices. SiC also conducts heat better than copper. Silicon wins on hole mobility, thermal conductivity (among the fast materials), oxide quality and cost.

1.3 Alloys and heterostructures

Mixing two compounds with a common element makes a ternary alloy, such as $Al_xGa_{1-x}As$ or $In_xGa_{1-x}As$. To first order its lattice constant varies linearly with composition (**Vegard's law**), and its band gap varies smoothly, usually with a slight downward **bowing**:

$$E_g(x) \approx (1-x)\,E_g^A + x\,E_g^B - b\,x(1-x)$$

Figure 4.1 is the map that compound-semiconductor engineers work from. A material can be grown on a substrate without defects only if their lattice constants are close. AlAs and GaAs differ by only 0.14%, so AlGaAs can be grown on GaAs at any composition: this is why GaAs/AlGaAs was the first heterostructure system. $In_{0.53}Ga_{0.47}As$ is matched to InP. A small mismatch can be accommodated by strain if the layer is thinner than a **critical thickness**; these strained **pseudomorphic** layers are used deliberately, as in the pHEMT (Section 3.3).

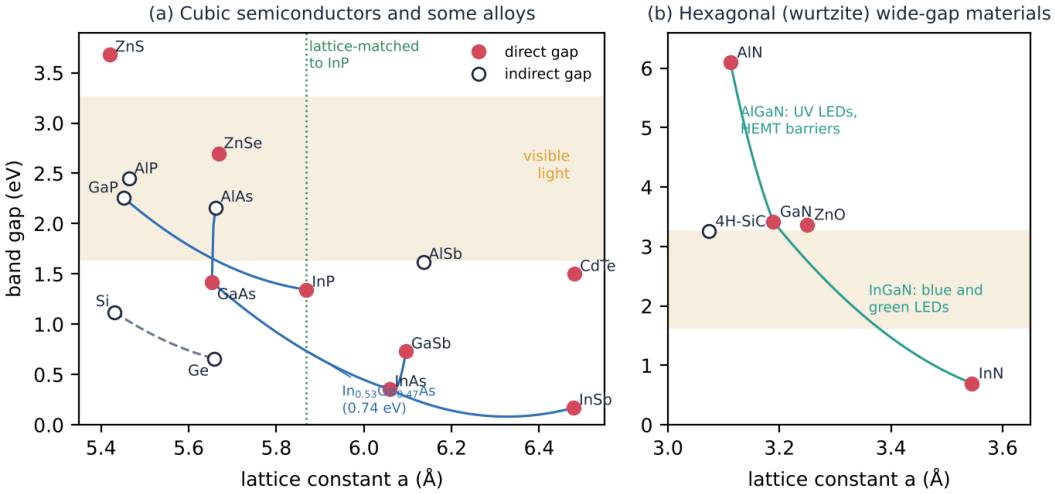


Figure 4.1. *(a) Band gap against lattice constant for cubic semiconductors. Filled circles: direct gap; open: indirect. Lines are alloys. The dotted line marks materials lattice-matched to InP. (b) The hexagonal nitrides and SiC: InGaN spans the visible spectrum (shaded) and AlGaN reaches the deep ultraviolet. Figure made for this course; values from the Ioffe database, bowing parameters from Vurgaftman et al. (2001).*

Check your understanding

1. Why can AlGaAs be grown on GaAs at any composition, but InAs not on GaAs?
2. Which composition of InGaAs is lattice-matched to InP, and what is its gap?

2 Bands: direct gaps, light electrons and the velocity peak

> **Key ideas**
>
> In a direct-gap material the bottom of the conduction band is directly above the top of the valence band, so an electron and a hole can recombine and emit a photon without help. In silicon they cannot.
>
> The conduction-band minimum of GaAs has a light mass ($0.067\ m_0$), which gives high mobility. A few tenths of an eV higher sit heavy "satellite" valleys.
>
> At about 3–4 kV/cm, hot electrons transfer to the satellite valleys and slow down: the velocity peaks and then falls. This is the transferred-electron (Gunn) effect.

2.1 Direct and indirect gaps

An electron and a hole can recombine and emit a photon only if momentum is conserved. A photon carries almost no momentum. In GaAs the conduction-band minimum and the valence-band maximum are both at Γ ($k = 0$), so the transition is **direct** and fast (Figure 4.2). In silicon the conduction-band minimum is near the X point, at the zone edge. Recombination needs a phonon to supply the momentum, a much slower, second-order process, and silicon is a poor light emitter. The emitted wavelength of a direct-gap material is $\lambda = hc/E_g$, so λ (nm) $\approx 1240/E_g$ (eV): 873 nm for GaAs, the near infrared.

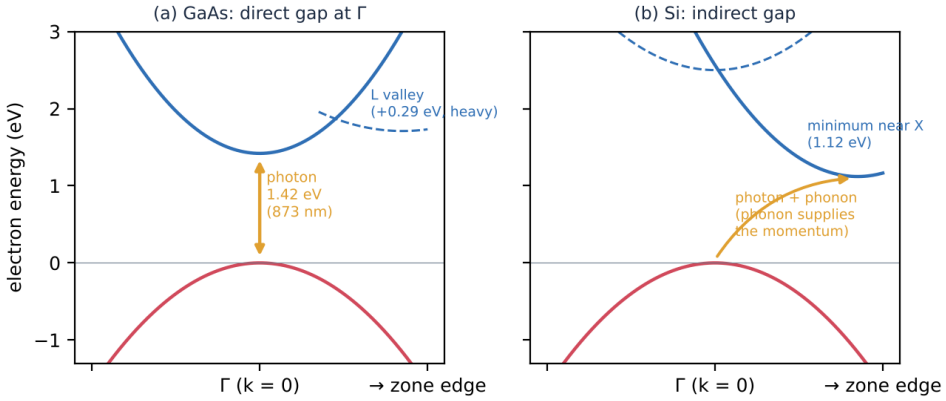


Figure 4.2. *Band structures near the gap. (a) GaAs: direct gap at Γ, with heavier L valleys about 0.29 eV higher. (b) Si: indirect gap; an optical transition needs a phonon. Figure made for this course; schematic.*

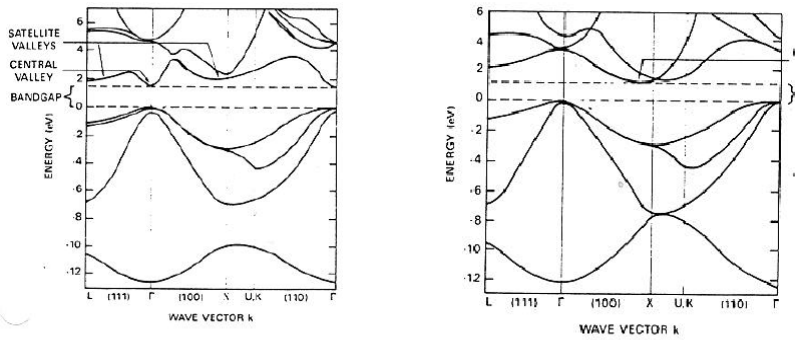


Figure 4.3. *Calculated band structures of GaAs and Si along the main symmetry directions, showing the direct gap of GaAs at Γ and the satellite valleys, and silicon's conduction-band minimum near X. Source: after J. R. Chelikowsky and M. L. Cohen, Phys. Rev. B 14, 556 (1976); reproduced from CSUN MSE-630 slides.*

2.2 Mass, mobility and valleys

The mobility is $\mu = q\tau/m^*$. The light Γ-valley mass of GaAs, 0.067 m_0, against a conductivity mass of about 0.26 m_0 for silicon electrons, accounts for most of GaAs's sixfold advantage in mobility. Holes are different: the valence bands of III–Vs are as heavy as silicon's or heavier, and hole mobilities are modest (400 $\mathrm{cm^2/V\,s}$ in GaAs). That asymmetry will matter in Section 7.

2.3 The transferred-electron effect

In GaAs the L valleys lie about 0.29 eV above the Γ minimum, with a much heavier mass and a much higher density of states. At low fields almost all electrons are in the Γ valley and move fast. As the field rises, electrons gain energy; at about 3–4 kV/cm enough of them reach 0.29 eV to scatter into the L valleys, where they are heavy and slow. The average velocity therefore **peaks**, at about 2×10^7 cm/s, and then **falls** towards about 1×10^7 cm/s (Figure 4.4). Over that range the differential mobility $\mathrm{d}v/\mathrm{d}E$ is negative.

This was predicted by Ridley, Watkins and Hilsum (1961–62) and observed by Gunn (1963) as microwave oscillations in a bar of GaAs: a **Gunn diode**. Negative differential mobility makes a uniform field unstable, and charge domains form and travel across the sample, oscillating at a frequency set by its length. Gunn diodes were the standard millimetre-wave sources for decades.

> **A correction to the old notes.** The CSUN deck states that "drift velocity slows in fields exceeding ~1000 V/cm". Its own plot shows the GaAs peak at about 3–4 kV/cm, as here. Below that field the velocity is still rising; the fall begins only once electrons transfer to the L valleys.

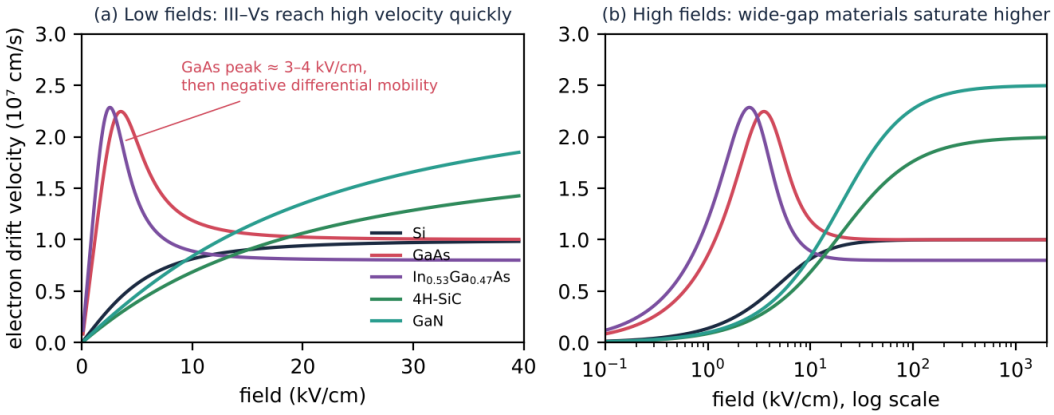


Figure 4.4. *Electron drift velocity against field. (a) Up to 40 kV/cm: GaAs and InGaAs reach about twice silicon's saturation velocity at a few kV/cm, then fall back. (b) On a log scale up to MV/cm: the wide-gap SiC and GaN saturate later but higher. Figure made for this course from standard empirical fits; the weak velocity peak of GaN is omitted.*

Two lessons for transistors follow from Figure 4.4. At the low fields of a long channel, III–V electrons are four to eight times faster than silicon's. At the high fields of a short channel, all the materials end up near 10^7 cm/s, but III–V electrons reach high velocity over a shorter distance, and in very short channels they can **overshoot** the steady-state curve before they scatter into the satellite valleys. Much of the speed advantage of III–V transistors comes from this.

Check your understanding

1. Why does silicon need a phonon to emit light?
2. Explain in three sentences why the GaAs velocity–field curve has a peak.

3 Heterojunctions and the HEMT

> **Key ideas**
>
> At a heterojunction the band-gap difference divides into a conduction-band step ΔE_c and a valence-band step ΔE_v.
>
> In a HEMT, electrons from donors in a wide-gap layer fall into the narrow-gap layer and form a two-dimensional electron gas (2DEG) away from the donors, so they move fast. The gate controls the 2DEG density.
>
> In AlGaN/GaN, the polarisation of the wurtzite crystals creates a 2DEG of about 10^{13} cm^{-2} with no doping at all: high current and high breakdown together.

3.1 Band alignment

When two semiconductors meet, their band gaps differ by ΔE_g, which is split between the two bands: $\Delta E_g = \Delta E_c + \Delta E_v$. For $Al_xGa_{1-x}As$ on GaAs, about 60–65% of the difference goes into the conduction band, so $\Delta E_c \approx 0.3$ eV for $x = 0.3$. When both steps confine electrons and holes in the same layer, the alignment is **type I** (the basis of the quantum-well laser of Lecture 2). When they confine electrons and holes in different layers, it is **type II**.

3.2 Modulation doping and the HEMT

Lecture 2 described the idea. Dingle and co-workers at Bell Labs (1978) doped only the AlGaAs layer of an AlGaAs/GaAs heterostructure. Electrons from the donors fall over the ΔE_c step into the undoped GaAs, leaving the ionised donors behind. The resulting field bends the bands into a triangular quantum well at the interface, and the electrons form a **two-dimensional electron gas (2DEG)** a few nanometres thick (Figure 4.5a). An undoped AlGaAs **spacer** a few nanometres thick separates the 2DEG further from the donors' Coulomb scattering. At low temperature the mobility of the 2DEG can exceed 10^6 cm^2/V s; at room temperature it is about 8,000 cm^2/V s, limited by phonons.

Mimura and co-workers at Fujitsu (1980) put a Schottky gate on the AlGaAs and made the **HEMT**. The gate controls the 2DEG density like the gate of a MOSFET, with the AlGaAs barrier playing the part of the oxide:

$$n_s \approx \frac{\varepsilon_b}{q\,(d + \Delta d)}\,(V_G - V_T)$$

where d is the barrier thickness and $\Delta d \approx 2$–4 nm is the distance of the 2DEG centroid from the interface (its quantum-capacitance correction, as for the CNT in Document 2).

> **Worked example: 2DEG density and current**
>
> AlGaAs barrier: $d = 25$ nm, $\varepsilon_b = 12.2\varepsilon_0$, $\Delta d = 3$ nm. Then $\varepsilon_b/q(d + \Delta d) = (12.2 \times 8.85 \times 10^{-14})/(1.6 \times 10^{-19} \times 2.8 \times 10^{-6}) = 2.4 \times 10^{12}$ cm^{-2} per volt. At 0.5 V of overdrive, $n_s = 1.2 \times 10^{12}$ cm^{-2}. If the electrons move at 2×10^7 cm/s, the current per width is $qn_sv = 1.6 \times 10^{-19} \times 1.2 \times 10^{12} \times 2 \times 10^7 = 3.8$ A/cm $= 0.38$ A/mm. GaAs HEMTs deliver a few hundred mA/mm, consistent with this.

3.3 pHEMTs and InP HEMTs

Replacing the GaAs channel with a thin, strained InGaAs layer (15–25% indium) on GaAs gives a deeper well, a lighter mass and higher mobility: the **pseudomorphic HEMT (pHEMT)**, the standard low-noise and switching transistor of the 1990s and 2000s. On InP substrates, InGaAs channels with 53–80% indium give the fastest transistors of any kind: in 2015 Northrop Grumman demonstrated amplification at 1 THz with a 25 nm InP HEMT, with f_{max} about 1.5 THz (Mei *et al.*, 2015). InP HEMTs are the amplifiers of choice for radio astronomy and sub-THz instruments.

3.4 GaN: electrons from polarisation

GaN and AlGaN are wurtzite crystals with a built-in electrical polarisation along the c axis (**spontaneous polarisation**). AlGaN grown on GaN is also strained, which adds a **piezoelectric** polarisation. At the AlGaN/GaN interface the polarisation changes abruptly, leaving a fixed sheet of positive charge σ_{pol} of order 10^{13} q per cm^2. Electrons gather at the interface to neutralise it, forming a 2DEG of about 10^{13} cm^{-2} **without any doping** (Ambacher *et al.*, 1999; Figure 4.5b). That is ten times the density of an AlGaAs/GaAs HEMT.

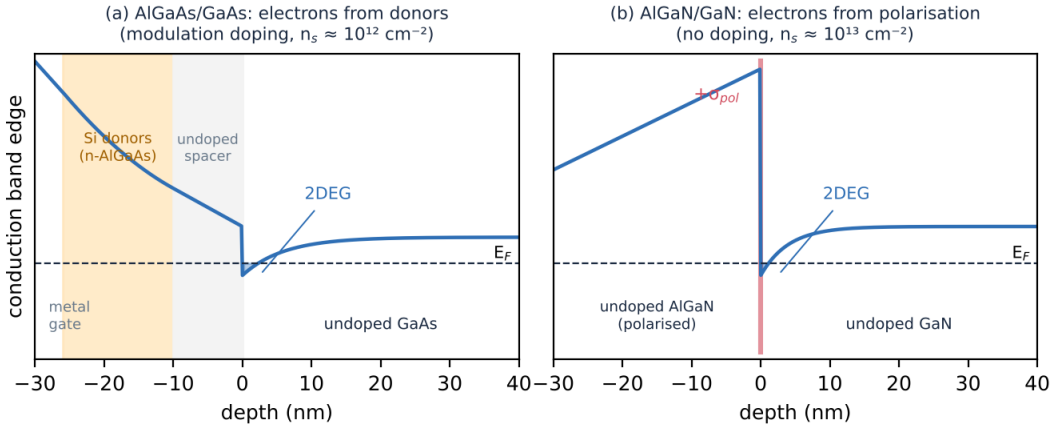


Figure 4.5. *Conduction-band edge of two HEMT structures. (a) AlGaAs/GaAs: electrons from Si donors in the AlGaAs fall into a triangular well at the interface, separated from the donors by an undoped spacer. (b) AlGaN/GaN: the polarisation charge at the interface (red) pulls the band below the Fermi level and holds a dense 2DEG with no doping. Figure made for this course; schematic.*

> **Worked example: why GaN HEMTs are power transistors**
>
> $n_s = 10^{13}$ cm^{-2} at $v = 1.5 \times 10^7$ cm/s gives $qn_sv = 1.6 \times 10^{-19} \times 10^{13} \times 1.5 \times 10^7 = 24$ A/cm $= 2.4$ A/mm, about six times the GaAs example. GaN also withstands about eight times the field, so the drain voltage can be several times higher. Power is roughly current × voltage, so the RF power per mm of gate width is ten times that of GaAs: 5–10 W/mm in production devices.

GaN HEMTs are normally **on**, because the 2DEG exists at zero gate bias. For power switches, which must be off when undriven for safety, a p-type GaN layer under the gate lifts the band and depletes the 2DEG (the "p-GaN gate" enhancement-mode device). The main reliability problems are **trapping** (electrons captured in the AlGaN surface or the GaN buffer temporarily reduce the current, called "current collapse") and the heat generated in a very small area. RF GaN is therefore grown on **SiC** substrates, which conduct heat well; power GaN is grown on cheaper **silicon** substrates.

Check your understanding

1. Where do the electrons in an AlGaN/GaN 2DEG come from?
2. Why is the GaN HEMT normally on, and how is it made normally off?

4 The heterojunction bipolar transistor

> **Key ideas**
>
> In a bipolar transistor, holes injected back from the base into the emitter waste current and limit the gain.
>
> A wide-gap emitter adds a valence-band step that blocks those holes. The base can then be doped very heavily, which lowers its resistance and raises the maximum frequency.
>
> InGaP/GaAs HBTs are the power amplifiers in almost every mobile phone; SiGe HBTs are silicon's version; InP HBTs pass 1 THz.

4.1 Kroemer's idea

In an npn transistor the emitter injects electrons into the base, and the base injects holes back into the emitter. Only the electrons are useful. In a homojunction transistor the ratio of the two currents is set by the doping ratio, so the emitter must be doped much more heavily than the base. But a lightly doped base has a high resistance, which limits speed.

Herbert Kroemer proposed in 1957 that a **wider-gap emitter** would solve this. The extra band gap appears mostly as a valence-band step ΔE_v at the emitter–base junction, which adds a barrier for holes but not for electrons (Figure 4.6). The hole back-injection is suppressed by roughly $\exp(-\Delta E_v/kT)$, a factor of 10^5 for $\Delta E_v = 0.3$ eV. The base can then be doped *more* heavily than the emitter, for a low base resistance and a thin base. Kroemer shared the 2000 Nobel Prize in Physics (with Zhores Alferov) for heterostructures.

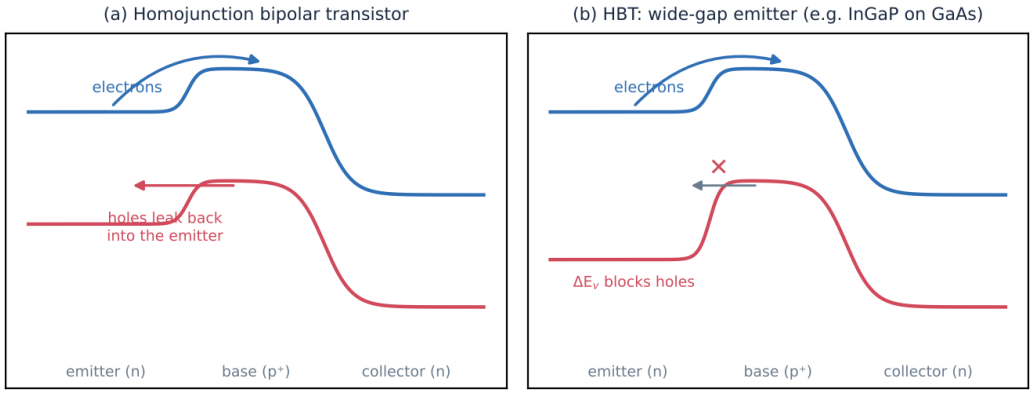


Figure 4.6. *Band diagrams of an npn transistor under forward bias. (a) Homojunction: holes from the base leak back into the emitter. (b) HBT with a wide-gap emitter: the valence-band step ΔE_v blocks them, while electrons still flow freely into the base. Figure made for this course; schematic.*

4.2 HBTs in use

- **InGaP/GaAs HBTs** (and some AlGaAs/GaAs) are the RF power amplifiers in almost every mobile phone, because they deliver high efficiency and linearity from a single battery-voltage supply.
- **SiGe HBTs** put a graded SiGe base (narrower gap) into a silicon process, giving the same benefit in reverse. SiGe BiCMOS is used for automotive radar at 77 GHz and optical-link circuits.
- **InP HBTs** reach f_{max} above 1 THz and drive the fastest optical-fibre transceivers.

The old notes give an MBE doping profile for an AlGaAs/GaAs HBT: an n-AlGaAs emitter at about 10^{17} cm^{-3}, a p^{+} GaAs base above 10^{19} cm^{-3}, and an n-GaAs collector at 10^{16} cm^{-3}. That base is doped a hundred times more heavily than the emitter, which a homojunction transistor cannot do.

Check your understanding

1. Why does a homojunction transistor need a more heavily doped emitter than base?
2. What does the heterojunction allow instead, and why does it raise f_{max}?

5 Wide-gap semiconductors for power

> **Key ideas**
>
> A power switch must block a high voltage when off and have a low resistance when on. For a given breakdown voltage, the ideal on-resistance falls as $1/(\varepsilon\mu E_c^3)$.
>
> SiC and GaN have critical fields about ten times silicon's, so their ideal on-resistance is several hundred times lower.
>
> SiC MOSFETs drive the motors of most new electric cars; GaN transistors are in small chargers and data-centre power supplies.

5.1 The Baliga figure of merit

A vertical power device blocks voltage across a lightly doped **drift layer**. To block V_B without exceeding the critical field E_c, the layer must be thick ($W \approx 2V_B/E_c$) and lightly doped ($N_D \approx \varepsilon E_c^2/2qV_B$). Its on-resistance per area is then $R = W/(q\mu N_D)$, which gives

$$R_{on,sp} = \frac{4V_B^2}{\varepsilon\,\mu\,E_c^3}$$

The denominator, $\varepsilon\mu E_c^3$, is the **Baliga figure of merit** (Baliga, 1982). Because E_c enters cubed, a tenfold increase in critical field beats a tenfold loss in mobility by a factor of a hundred.

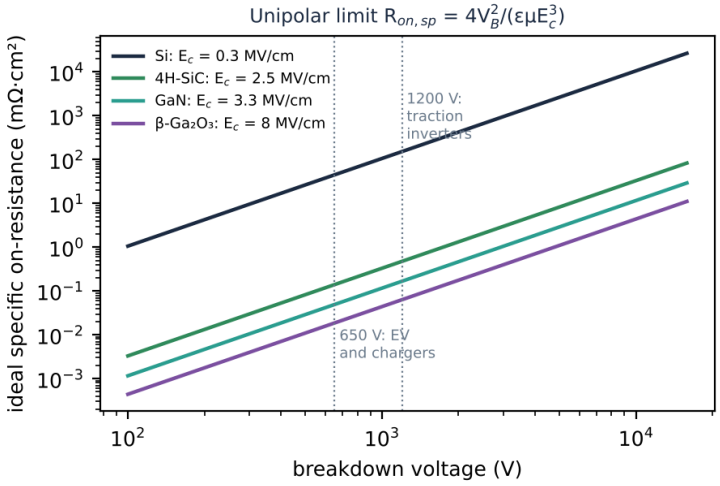


Figure 4.7. The ideal (unipolar) limit of on-resistance against breakdown voltage for Si, 4H-SiC, GaN and β-Ga_2O_3. Figure made for this course from the formula above with the table values.

Worked example: a 600 V switch

Silicon ($\varepsilon = 11.7\varepsilon_0$, $\mu = 1350$ cm^2/V s, $E_c = 0.3$ MV/cm): $R_{on,sp} = 4 \times 600^2 / (1.04 \times 10^{-12} \times 1350 \times 2.7 \times 10^{16}) = 0.038\ \Omega\ \text{cm}^2$. GaN ($9\varepsilon_0$, 1200 cm^2/V s, 3.3 MV/cm) gives $4.2 \times 10^{-5}\ \Omega\ \text{cm}^2$, about 900 times less. A switch of 50 mΩ needs 0.8 cm^2 of silicon drift region but under 0.1 mm^2 of GaN. In practice silicon gets closer to its limit with superjunction structures, and SiC and GaN fall short of theirs, but the advantage remains large. Smaller chips mean lower capacitance and faster switching, so the magnetics and capacitors around the switch shrink too. That is why a 65 W GaN charger can be smaller than an older silicon charger of a fraction of the power.

5.2 SiC and GaN in 2026

- **SiC MOSFETs** at 650–1200 V are used in electric-vehicle traction inverters (first in high volume in the Tesla Model 3, from 2018), on-board chargers, solar inverters and industrial drives. Wafers moved from 150 mm to 200 mm in the 2020s. The main device challenge is the SiC/SiO_2 interface, whose high trap density lowers channel mobility far below the bulk value.
- **GaN-on-silicon HEMTs** at 100–650 V are used in USB-C chargers, data-centre and telecom power supplies, and lidar drivers. They are lateral devices grown on 150–200 mm silicon. In 2024 Infineon announced the first 300 mm GaN power wafers, with about 2.3 times as many chips per wafer as 200 mm and samples due at the end of 2025.
- **Gallium oxide** (β-Ga_2O_3, gap about 4.8 eV, E_c about 8 MV/cm) is the next candidate. It can be grown from the melt, like silicon, but it has no usable p-type doping and poor thermal conductivity.

Check your understanding

1. Derive $R_{on,sp} = 4V_B^2/\varepsilon\mu E_c^3$ from the drift-layer thickness and doping.
2. Why does a GaN charger need smaller inductors and capacitors than a silicon one?

6 Making compound devices

Key ideas

Compound wafers are smaller, more brittle and far more expensive than silicon; GaN is usually grown on a foreign substrate.

Heterostructures are grown by molecular-beam epitaxy (MBE) or metal-organic chemical vapour deposition (MOCVD), with control to a single atomic layer.

GaAs has no useful native oxide, and arsenic leaves the surface above about 500 °C. Processes use Schottky gates, deposited dielectrics, alloyed ohmic contacts and implant isolation.

6.1 Substrates

- **GaAs**: semi-insulating wafers 150 mm across are standard (the CSUN deck's 4-inch, $300 wafers are a 1990s figure). Grown by the vertical-gradient-freeze or liquid-encapsulated Czochralski methods; more brittle than silicon.
- **InP**: 100–150 mm, expensive and fragile.

- **SiC**: 150 mm moving to 200 mm. Growth by sublimation at about 2300 °C is slow, and SiC wafers remain much dearer than silicon.
- **GaN**: bulk GaN wafers exist but are small and costly. Most GaN is grown on SiC (RF), on silicon (power) or on sapphire (LEDs), with buffer layers to manage the lattice and thermal-expansion mismatch.

6.2 Epitaxy

Molecular-beam epitaxy (Cho and Arthur, Bell Labs, 1970s) evaporates the elements from heated cells in ultra-high vacuum onto a heated substrate (Figure 4.8). Shutters switch each beam on and off in a fraction of a second, so layers can be changed one atomic plane at a time, and electron diffraction monitors the growth as it happens. MBE grew the first HEMTs and most research heterostructures. **MOCVD** (Manasevit, 1968) uses metal-organic gases such as trimethylgallium with arsine or ammonia at atmospheric or reduced pressure. It is faster and scales to many large wafers at once, and it is how almost all LEDs, lasers, HBTs and GaN devices are made today.

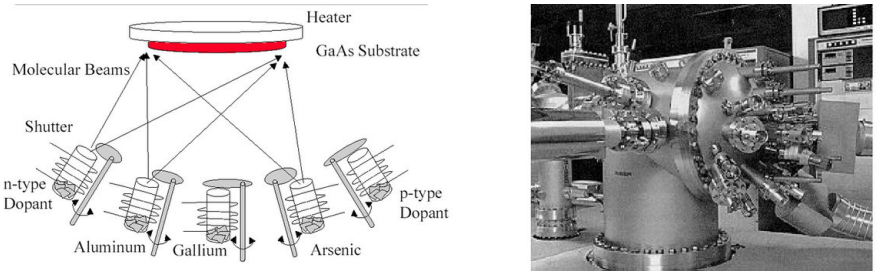


Figure 4.8. Molecular-beam epitaxy: schematic of the beams from the effusion cells onto the heated substrate (left) and a production MBE system (right). Reproduced from CSUN MSE-630 slides.

6.3 Processing without an oxide

The CSUN deck lists the practical consequences, which are unchanged:

- **No native oxide.** The oxides of GaAs (Ga_2O_3, As_2O_3, As_2O_5) are leaky and full of traps that pin the surface Fermi level. So GaAs transistors used Schottky-barrier metal gates (the **MESFET**) and later heterojunction barriers (the HEMT). Deposited oxides (Al_2O_3 by ALD) work on InGaAs, but with higher interface-trap densities than SiO_2 on silicon.
- **Arsenic loss.** Above about 500 °C arsenic leaves the surface, so anneals are done with a SiO_2 or Si_3N_4 cap or in an arsenic overpressure.
- **Ohmic contacts.** Au–Ge alloyed above the Au–Ge eutectic (361 °C; the CSUN deck says 356 °C), with a Ni layer, dopes the GaAs surface n^+ under the contact. Schottky contacts use metals such as Ti/Pt/Au or Al.
- **Isolation.** Ion implantation damages the material outside the devices, making it semi-insulating; there is no LOCOS or STI.

The CSUN deck's ten-step GaAs MESFET process (channel-stop implant, channel etch, gate lift-off, self-aligned source and drain implant, anneal under an encapsulant, ohmic metal) remains a good picture of a compound process, which uses far fewer, simpler steps than advanced silicon CMOS.

Check your understanding

1. Why did GaAs transistors use Schottky gates rather than MOS gates?
2. Why is MOCVD, not MBE, used for mass production of LEDs?

7 Why compound semiconductors are not in CMOS logic

> **Key ideas**
>
> High electron mobility does not translate into a better logic transistor. A light mass also means a small density of states, so the channel holds less charge per volt of gate voltage: the density-of-states bottleneck.
>
> III–Vs have slow holes, poor oxides, small gaps (leakage) and are hard to integrate on silicon.
>
> Silicon kept improving through strain, high-κ, FinFETs and nanosheets. III–Vs are used on silicon for RF and light, not for logic.

7.1 The density-of-states bottleneck

For a decade (about 2005–2015) Intel and others built InGaAs quantum-well transistors as possible n-channel replacements for silicon, with germanium for the p-channel. They switched well, but they did not deliver the expected current. The reason is the quantum capacitance of Document 2, in two dimensions.

A two-dimensional electron gas has a density of states $D_{2D} = g_v m^*/\pi\hbar^2$ per unit area (spin included), so its quantum capacitance is

$$C_Q = q^2 D_{2D} = \frac{q^2 g_v m^*}{\pi\hbar^2}$$

A light mass, the very thing that gives high mobility, gives a small C_Q, and the gate charge $C_G V$ is reduced by the series combination with the oxide capacitance. Solomon and Laux (2001) called this the **density-of-states bottleneck**.

> **Worked example: InGaAs against silicon**
>
> For one valley with $m^* = 0.067\, m_0$: $C_Q = (1.6 \times 10^{-19})^2 \times 0.067 \times 9.11 \times 10^{-31} / (\pi \times (1.055 \times 10^{-34})^2) = 0.045$ F/m^2 $= 4.5$ μF/cm^2. With an oxide of 1 nm EOT ($C_{ox} = 3.45$ μF/cm^2), the gate capacitance is $(1/3.45 + 1/4.5)^{-1} = 1.95$ μF/cm^2: the channel loses 44% of the charge. For a silicon (100) inversion layer, two valleys with $m^* = 0.19\, m_0$ give $C_Q \approx 25$ μF/cm^2 and $C_G = 3.0$ μF/cm^2. The III–V's higher injection velocity is largely cancelled by its lower charge.

7.2 The other obstacles

- **Holes.** III–V hole mobilities are no better than silicon's, so CMOS would need a second material (germanium) for the p-channel.
- **Gate stack.** Interface-trap densities on InGaAs remain well above those on silicon, degrading the subthreshold swing and reliability.
- **Small gaps.** $In_{0.53}Ga_{0.47}As$ has a gap of 0.74 eV, and band-to-band tunnelling at the drain raises the off-current in short channels.

- **Integration.** Growing III–Vs on silicon creates dislocations from the lattice mismatch (4% for GaAs, 8% for InP) and antiphase domains, and every extra material adds process cost.
- **A moving target.** Strained silicon, high-κ/metal gates, FinFETs and nanosheets kept silicon ahead, as Lecture 1 described.

7.3 Where III–Vs meet silicon instead

The III–V-on-silicon work has been redirected to things silicon cannot do. GaN on silicon makes power devices (Section 5) and RF front ends; imec and others are growing InP HBTs on 300 mm silicon for sub-THz radios. III–V lasers are bonded onto silicon-photonics (SOI) wafers to make the light sources that silicon lacks (Document 3, Section 8). This is heterogeneous integration: each material used for what it does best.

Check your understanding

1. Why does a light effective mass reduce the gate charge of a transistor?
2. List three reasons, other than the density-of-states bottleneck, why III–Vs are not used for CMOS logic.

8 Where compound semiconductors are used

> **Key ideas**
>
> Compound semiconductors own the places silicon cannot reach: light emission, the highest frequencies, high power and high voltage.
>
> Gallium is a by-product of aluminium and zinc refining, and most of the world's primary supply comes from China, which has restricted exports since 2023.

Material	Main uses in 2026
GaAs	Phone power amplifiers (HBT) and switches (pHEMT, now largely RF-SOI), VCSELs for face recognition and data links, red and infrared LEDs, space solar cells
InP	Lasers and photodiodes for optical fibre at 1.3 and 1.55 μm, the fastest optical transceivers, sub-THz amplifiers
GaN	Blue, white and green LEDs and blue lasers (Lecture 2), radar and 5G base-station amplifiers, chargers and power supplies
SiC	EV traction inverters, chargers, solar and industrial power; substrates for RF GaN
InGaAs / InSb / HgCdTe	Infrared cameras and detectors

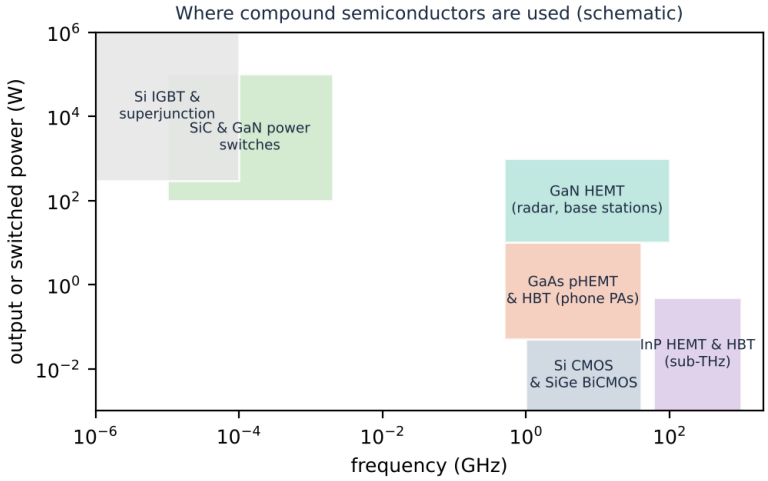


Figure 4.9. *Where each technology is used, by frequency and power. Figure made for this course; schematic boundaries.*

8.1 Supply and safety

The CSUN deck says gallium "is rarer than gold". It is not. Gallium makes up about 19 parts per million of the Earth's crust, comparable to lead and thousands of times more than gold (about 0.004 ppm). It is scarce in another sense: it is not mined for itself but recovered as a by-product of refining bauxite into alumina and of zinc processing, so supply depends on those industries. China produces about 98% of the world's primary gallium. It imposed export licensing on gallium and germanium in August 2023 and banned their export to the United States in December 2024. Supply security is now part of the compound-semiconductor story.

Arsenic and its compounds are toxic, and the arsine and phosphine gases used in MOCVD are extremely toxic. Compound-semiconductor fabs handle them with gas cabinets, monitoring and abatement. GaAs waste and polishing slurries must be treated as hazardous.

Check your understanding

1. Name one application each for GaAs, InP, GaN and SiC, and say which property it relies on.
2. Why is gallium supply a concern although gallium is not rare?

9 Summary

- III–V, II–VI and IV–IV compounds can be **alloyed** and grown as **heterostructures**; Figure 4.1 is the design map.
- **Direct gaps** emit light ($\lambda \approx 1240/E_g$ nm); silicon's indirect gap does not. A light Γ-valley mass gives GaAs and InGaAs high mobility, and the transferred-electron effect gives a velocity peak of about 2×10^7 cm/s at 3–4 kV/cm.
- **HEMTs** confine a 2DEG at a heterojunction away from its donors, $n_s \approx \varepsilon(V_G - V_T)/q(d + \Delta d)$. pHEMTs and InP HEMTs give the lowest noise and highest frequencies (1 THz). **AlGaN/GaN** HEMTs get 10^{13} cm^{-2} from polarisation and deliver several A/mm and 5–10 W/mm.

- **HBTs** use a wide-gap emitter to block hole back-injection, allowing a heavily doped base: phone PAs (InGaP/GaAs), SiGe BiCMOS, InP above 1 THz.
- **Power**: $R_{on,sp} = 4V_B^2/\varepsilon\mu E_c^3$. SiC and GaN beat silicon by hundreds of times in principle and now dominate EV inverters and fast chargers.
- Compound processing differs: smaller wafers, MBE and MOCVD, no native oxide, arsenic loss, alloyed contacts, implant isolation.
- III–Vs are **not** used for logic: density-of-states bottleneck, slow holes, poor gate stacks, small gaps, integration. They are integrated with silicon for RF, power and light instead.

Corrections to the old notes

Old note	What it said	Correction
CSUN MSE-630	"Gallium … is rarer than gold"	About 19 ppm of the crust, against about 0.004 ppm for gold; scarce because it is only a by-product
CSUN MSE-630	"Drift velocity slows in fields exceeding ~1000 V/cm"	The GaAs peak is at about 3–4 kV/cm, as the deck's own plot shows
CSUN MSE-630	"The main semiconductor device made in GaAs is the MESFET"	Today it is the HBT (phone PAs) and the pHEMT; GaN HEMTs lead in RF power
CSUN MSE-630	"Small size (4") ingots"; "4" GaAs wafers cost about $300"	150 mm GaAs is standard; prices are a 1990s figure
CSUN MSE-630	Au–Ge eutectic at 356 °C	361 °C
CSUN MSE-630	Covers GaAs only	GaN, SiC and InP are now at least as important (Sections 3–6)

Words to know

Term	Meaning
Heterostructure	Layers of different semiconductors grown on one another as one crystal
Vegard's law	Linear variation of an alloy's lattice constant with composition
Direct / indirect gap	Conduction-band minimum at the same / a different k as the valence-band maximum
Transferred-electron effect	Velocity drop as hot electrons move to heavy satellite valleys (Gunn effect)
2DEG	Two-dimensional electron gas at a heterojunction
HEMT, pHEMT	High-electron-mobility transistor; with a strained (pseudomorphic) channel
Polarisation charge	Fixed interface charge from the spontaneous and piezoelectric polarisation of wurtzite crystals
HBT	Heterojunction bipolar transistor, with a wide-gap emitter
Baliga figure of merit	$\varepsilon\mu E_c^3$, the material merit of a unipolar power device
MBE, MOCVD	Molecular-beam epitaxy; metal-organic chemical vapour deposition

Term	Meaning
Density-of-states bottleneck	Low channel charge in light-mass materials, through their small quantum capacitance

In-class animation

Compound semiconductors (faizverse.github.io/nanoelectronic-devices/L4). Two tabs. *Velocity–field*: choose materials and compare their curves on linear and log scales; read off the low-field mobility, peak and saturation velocity. *Band-gap map*: move along an alloy line and read the band gap, lattice constant, mismatch to GaAs or InP, and emission wavelength.

Sources and further reading

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